Statistical Physics-Where Worlds Collide

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∞ Let Infinity Be Your Limit ∞

1 Introduction

Statistical physics is where mathematics stops being abstract and begins to describe reality at an astonishing scale. It is a subject where combinatorics, probability theory, and calculus orchestrate the behavior of matter itself.

At first glance, it may seem strange that randomness could produce precise physical laws. Yet, it is precisely through counting and probability that order emerges from chaos.

2 A Thought Experiment

Suppose, someone hands you 1000 coins. You flip them all at once. What might be the result you can expect? 500 might be a reasonable answer. But, what is the probability of getting exactly 500 heads or tails?As surprising as it may sound, it is just 2.5%. But, on the basis of our general "common-sense" it feels like there shall be much much more possibility for getting 500 heads than just mere 2.5% The key formula governing this scenario is indeed beautiful, as shown below:

$$\Omega(k) = \binom{1000}{k}$$

Here,

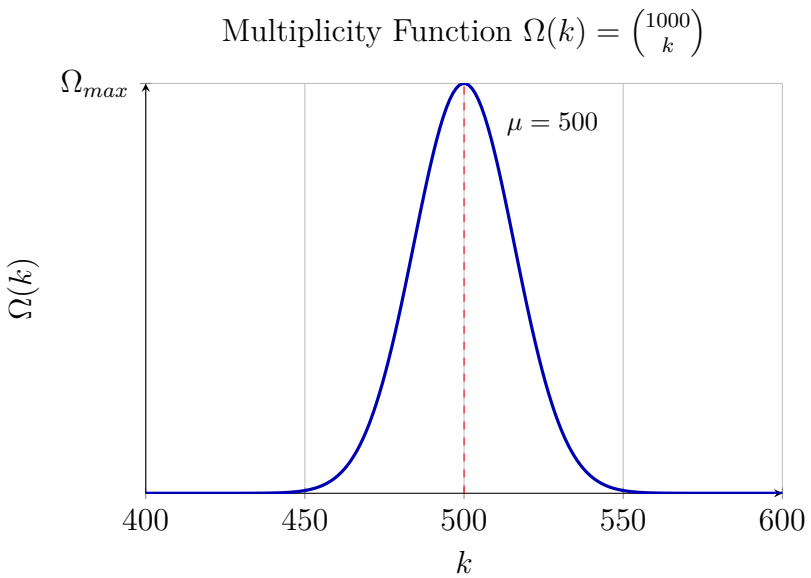
- As seen from the equation, There's only one way to get 0.
- There's astronomically many ways to get 500.

Nature doesn't prefer any orientation of order or disorder. It simply favors what is numerically overwhelming. Replace the coins with molecules and it turns into the basis of thermodynamics.

3 The Tyranny of Large Numbers

Let us zoom in on the function from the previous section.

$$\Omega(k) = \binom{1000}{k}$$



As seen, the graph is symmetric about k=500. It peaks at 500 and then falls off rapidly, too! But how rapidly does it fall off?For extremely large numbers, the factorial grows rapidly. Even for our 1000, the factorial is just something of an astronomical quantity. Just imagine how much larger it would be for something like $10^{23}, 10^{25}, 10^{100}$ and such orders of magnitude! But who would even realize seeing these that for such numbers, it gradually becomes **deterministic**.

In general, we can state the function $\Omega(k)$ as the ***multiplicity function*** which is described as thee follows:

$$\Omega(k) = \binom{N}{k}$$

where N is the number of molecules we have with us. Also, another helpful thing is that for enormous N, the binomial distribution becomes well approximated by a Gaussian Bell curve.

Stirling's Approximation: A Calculus Shortcut

We can see that to calculate the function $\Omega(k)$, we have to calculate really large factorials. To handle the factorials in $\binom{N}{k}$, we rely on a method called Stirling's Approximation. The derivation is a treat for calculus lovers. Consider $\ln(n!) = \sum_{x=1}^{n} \ln x$. For large n, we can approximate this sum as an integral:

$$\ln(n!) \approx \int_1^n \ln x \, dx$$

Using integration by parts ($u = \ln x, dv = dx$), we find:

$$[x \ln x - x]_1^n = n \ln n - n + 1$$

For $n \to \infty$, this gives us the famous $n \ln n - n$. This result is the "engine" that allows us to turn discrete counting into smooth, differentiable curves.

Now, we can try something, that is, we can try to approximate $\binom{N}{k}$ at around $k = \frac{N}{2}$. We can show that(the reader can try out the algebra):

$$\binom{N}{N/2} \approx \frac{2^N}{\sqrt{\pi N/2}}$$

This says something really useful and elegant. That is, if the total number of outcomes is 2^N, then the vast majority clusters around near $N/2$.We can now try out N=1000 for our example(Again, the reader is recommended to do the algebra).

For $N = 1000$, vast majority of the data clusters around 500 as given by the formula earlier. Also, the relative fluctuation size is about:

$$\frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}}$$

Which when calculated numerically for N=1000, gives around 3.16%.

Now, let's move on to the scaling law that changes our view completely!

$$\text{Fluctuation size} \sim \sqrt{N}$$

$$\text{Relative fluctuation} \sim \frac{1}{\sqrt{N}}$$

Now, if $N = 10^{23}$, the relative fluctuation becomes just about 10^{-11}, which is just almost insignificant! The graph of this distribution just looks like a needle and when viewed, looks just like a vertical straight line. This is often called the Thermodynamic limit.

We can also find the probability distribution function that gives me the probability for any k. The probability distribution is given by the formula:

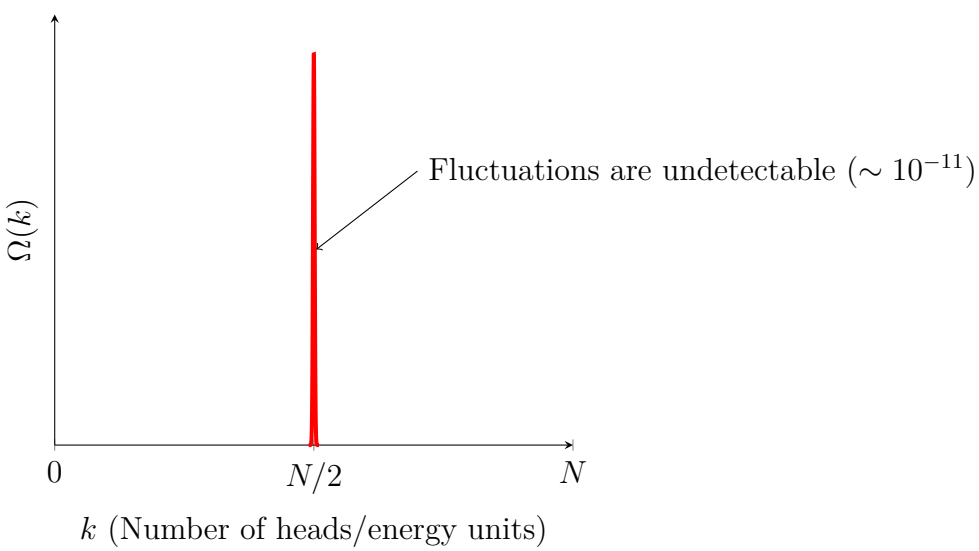
$$P(k) = \binom{N}{k}\left(\frac{1}{2}\right)^N$$

For $N = 1000$ and $k = 500$:

$$P(500) = \binom{1000}{500}(\frac{1}{2})^{1000} \approx 0.025225$$

Which is what we got initially. With this formula, we can get the probability of finding such microstates with k molecules out of all the possible microstates in existence.

Multiplicity in the Thermodynamic Limit



For enormous numbers like the one shown above, the graph has almost no distribution at all. Due to this sort of behavior of the curve we don't observe weird behaviors at the macroscopic scale. For example, we never observe all the air inside the room clustering to just one side of the room and making us gasp for oxygen!

4 The Physical Reality: Microstates & Macrostates

First of all, what even is a microstate or a macrostate?
Microstate is a specific, detailed configuration of all individual particles in a system (positions and momenta), while a **Macrostate** is the overall, bulk state defined by macroscopic properties like pressure, volume, and temperature. In statistical physics, the distinction between microstates and macrostates is where abstract combinatorics transforms into the tangible laws of our universe. A microstate is a specific, microscopic configuration of a system—a detailed "snapshot" specifying the exact state of every single molecule. In contrast, a macrostate is defined by global observables like temperature, pressure, or total energy. Macostate is what we experience all the time while all the microstates accumulate to give us that especific macrostate.

The bridge between these two world is the multiplicity function $\Omega(K)$, which counts the number of microstates that correspond to a single macrostate. In a system of N particles, the macrostate representing "equilibrium" condition (a 50/50 distribution) possesses a staggering number of microstates compared to any other "ordered" state.

As N grows toward the scale of Avogadro's number (10^{23}) , this statistical preference becomes an absolute mandate. The probability distribution narrows so sharply into a "needle" that the fluctuations away from the peak become effectively impossible to observe. This is the "collision" of worlds: what begins as a simple game of chance becomes, through the sheer weight of numbers, a deterministic physical law.

At the fundamental level, nature is a radical democrat. In a system of N binary particles (such as spins or coins), every single specific sequence—every **microstate** ϕ_i—is assigned the exact same probability:

$$P(\phi_i) = \frac{1}{\Omega_{\text{total}}}$$

There is no "force" pushing atoms toward equilibrium; a microstate where every atom is huddled in one corner is mathematically just as likely as any specific "messy" microstate. To understand why the total number of states is $\Omega_{\text{total}} = 2^N$, we simply apply the fundamental counting principle. If each of the N particles has 2 independent choices (Up or Down), the total combinations are:

$$\underbrace{2 \times 2 \times 2 \times \cdots \times 2}_{N \text{ times}} = 2^N$$

The "collision" occurs when we group these microstates into **macrostates** M_k, defined by a single observable k (the number of "up" spins). The probability of a macrostate is determined by its **multiplicity** $\Omega(k)$, which counts how many ways we can choose k positions out of N:

$$P(M_k) = \frac{\Omega(k)}{\Omega_{\text{total}}} = \frac{1}{2^N} \frac{N!}{k!(N-k)!}$$

While the individual microstates are equal, the macrostates are not. The "ordered" macrostate ($k = 0$) has a multiplicity of exactly 1, while the "disordered" equilibrium ($k = N/2$) has a multiplicity that scales factorially.

In the abstract world, $1/\sqrt{N}$ is just a small coefficient. In the physical world, for $N = 10^{23}$, it is the reason the Second Law of Thermodynamics is never broken. The "needle" is so sharp that the probability of even a 10^{-8} relative fluctuation away from equilibrium is smaller than the probability of winning every lottery on Earth simultaneously for a billion years.

5 Entropy and Boltzmann's Tombstone

In the framework of statistical mechanics, **Entropy** (S) is defined as a logarithmic measure of the number of microscopic configurations (Ω) that correspond to a specific macroscopic state. Centrally, it is expressed by the **Boltzmann Principle**:

$$S = k_B \ln \Omega$$

where k_B is the Boltzmann constant and Ω is the **multiplicity** (or statistical weight) of the macrostate.

For a more generalized system where each microstate i may have a different probability P_i, the definition expands into the **Gibbs Entropy** formula:

$$S = -k_B \sum_i P_i \ln P_i$$

This formulation reveals that entropy is essentially the *expected value* of the information content of the system. In the special case where all microstates are equally likely ($P_i = 1/\Omega$), the Gibbs formula collapses back into Boltzmann's original logarithmic definition.

The Logarithmic Mandate

If physics has a "sacred" equation, it is the one carved into Ludwig Boltzmann's gravestone in Vienna: $S = k_B \ln \Omega$. This elegant expression is the ultimate bridge where abstract counting meets physical reality. Here, k_B is the Boltzmann constant (1.38×10^{-23} J/K), acting as a scaling factor to translate the dimensionless units of information into the units of thermodynamics.

A mathematically inclined observer might ask: why a logarithm? The answer lies in the property of **additivity**. Consider two independent systems, A and B. In the world of combinatorics, if system A has Ω_A microstates and system B has Ω_B microstates, the total multiplicity of the combined system is multiplicative:

$$\Omega_{total} = \Omega_A \times \Omega_B$$

However, in the world of physics, we require entropy S to be an **extensive property**—meaning the total entropy of two blocks of metal should be the sum of their individual entropies ($S_{total} = S_A + S_B$). To transform a multiplicative relationship into an additive one, the only continuous functional solution is the logarithm:

$$f(\Omega_A \times \Omega_B) = f(\Omega_A) + f(\Omega_B) \implies f(\Omega) \propto \ln \Omega$$

Thus, the logarithm is a mathematical necessity to ensure that entropy behaves like mass or energy.

The Fundamentals of Statistical Weight

At its core, entropy measures the "spread" of a system across its available phase space. By applying **Stirling's Approximation** to the multiplicity of a large system, we can derive the entropy of an ideal system. For N particles with k units of energy, the entropy is proportional to the volume of the accessible microstates.

The Second Law of Thermodynamics is therefore reframed as a statistical certainty: systems evolve toward macrostates of higher entropy simply because those states contain the vast majority of the total available microstates. In the thermodynamic limit ($N \to 10^{23}$), the "most probable state" is so overwhelmingly dominant that the probability of observing a decrease in entropy is effectively zero, creating the macroscopic "Arrow of Time" from the microscopic laws of chance.

6 The Bridge: Maximizing Chaos under Constraints

We have established that entropy $S = k_B \ln \Omega$ measures the "multiplicity" of a state. If the second law of thermodynamics states that entropy must increase, it implies that a system will naturally evolve toward the **most probable macrostate**—the one with the largest Ω.

But for a gas in a container, this "search for chaos" is not unrestricted. The system is bound by two rigid physical constraints:

1. **Conservation of Particles:** The total number of molecules N is constant ($\sum n_i = N$).

2. **Conservation of Energy:** The total energy E of the system is fixed ($\sum n_i \epsilon_i = E$).

The Lagrange Multiplier Collision

To a mathematician, this is a classic **constrained optimization problem**. We want to maximize the function $\ln \Omega(n_1, n_2, \dots)$ subject to our two constraints. Using the method of **Lagrange Multipliers**, we define a functional:

$$\mathcal{L} = \ln \Omega + \alpha \left(N - \sum n_i \right) + \beta \left(E - \sum n_i \epsilon_i \right)$$

By setting the derivative $\frac{\partial \mathcal{L}}{\partial n_i} = 0$ and applying Stirling's Approximation, a stunning mathematical result emerges. The most probable distribution of particles across energy levels is not uniform; it is **exponential**:

$$n_i = C e^{-\beta \epsilon_i}$$

In this "collision" of calculus and counting, the multiplier β reveals itself to be the physical temperature ($\beta = 1/k_B T$).

This exponential term, known as the **Boltzmann Factor**, is the DNA of the Maxwell-Boltzmann distribution. It tells us that while nature wants to spread energy out to increase entropy, the requirement of a fixed total energy forces a "statistical competition." The result of this competition is the specific curve of speeds we see in every breath of air we take.

7 The Maxwell-Boltzmann Distribution: The Most Probable Chaos

Now, let's shift our focus to a container of gas at temperature T, where not every molecule moves at the same speed. Some are sluggish, while others are incredibly energetic. The **Maxwell-Boltzmann Distribution** describes the probability $P(v)$ of finding a particle with a specific velocity v.

The Exponential Decay of Energy

The probability that a particle occupies a state with energy ϵ is given by the **Boltzmann Factor**:

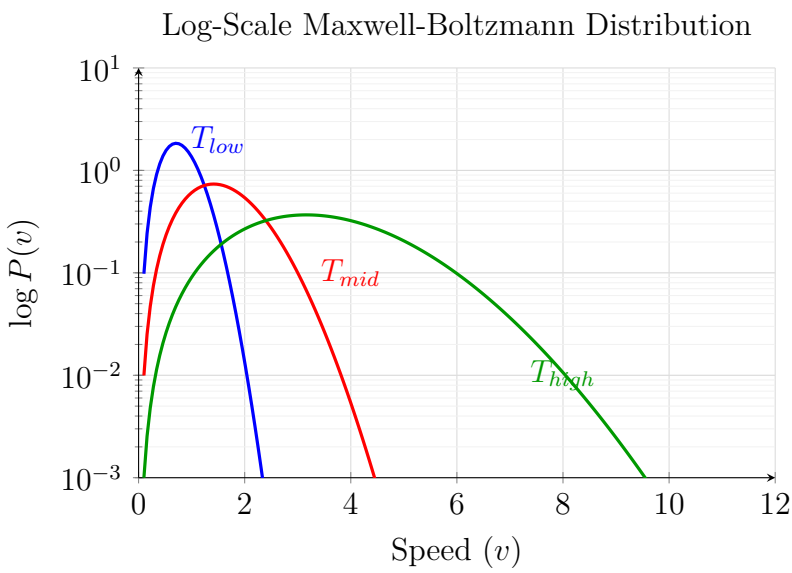
$$P(\epsilon) \propto e^{-\epsilon/k_B T}$$

This exponential decay is a mathematical consequence of maximizing the entropy of the system under the constraint of a fixed total energy. To a mathematician, this is an application of **Lagrange Multipliers**: we seek the distribution that maximizes $\ln \Omega$ while keeping $\sum n_i = N$ and $\sum n_i \epsilon_i = E_{total}$.

The Speed Distribution

While the probability of an individual state decreases exponentially with energy, the number of available states (the density of states) increases with velocity. In three-dimensional space, the "volume" of possible velocity vectors for a speed v is proportional to the surface area of a sphere, $4\pi v^2$. Multiplying the Boltzmann factor by this geometric factor gives us the **Maxwell-Boltzmann Speed Distribution**:

$$f(v) = \left(\frac{m}{2\pi k_B T}\right)^{3/2} 4\pi v^2 e^{-\frac{mv^2}{2k_B T}}$$



Interpreting the Curve

This formula reveals a beautiful competition between two mathematical forces:

- **The Quadratic Term (v^2):** Represents the increasing number of ways to have a high speed (geometric degeneracy).
- **The Exponential Term (e^{-v^2}):** Represents the decreasing probability of having high energy (statistical cost).

The "collision" of these two terms creates a skewed bell curve with a peak at the *most probable speed*. Even though the particles are colliding and changing speeds millions of times per second, the **overall shape** of this distribution remains rock-steady. This is the definition of thermal equilibrium: microscopic chaos yielding a perfectly predictable macroscopic curve.

8 Statistical Landmarks: 3 Ways to Measure Speed

The Maxwell-Boltzmann distribution $f(v)$ is a probability density function. To find the specific "speeds" that characterize a gas, we must apply the tools of calculus to the distribution function:

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}}$$

For simplicity in our derivations, let us define a constant $A = \frac{m}{2k_B T}$. The distribution then simplifies to $f(v) \propto v^2 e^{-Av^2}$.

1. Most Probable Speed (v_{mp})

The most probable speed is the **mode** of the distribution—the speed at the very peak of the curve. To find it, we set the first derivative of $f(v)$ to zero:

$$\frac{d}{dv}\left(v^2 e^{-Av^2}\right) = 0$$

Applying the product rule:

$$2ve^{-Av^2} + v^2(-2Av)e^{-Av^2} = 0$$

Factoring out the exponential (which is never zero), we get $2v - 2Av^3 = 0$. Solving for v (ignoring the trivial $v = 0$):

$$v_{mp} = \sqrt{\frac{1}{A}} = \sqrt{\frac{2k_B T}{m}}$$

2. Average Speed ($\langle v \rangle$)

The average (mean) speed is the expected value of v. Mathematically, this is the first moment of the distribution:

$$\langle v \rangle = \int_0^\infty v f(v)\, dv$$

This integral requires the use of the standard Gaussian integral form $\int_0^\infty x^3 e^{-ax^2} dx = \frac{1}{2a^2}$. After substituting our constants and evaluating, we find:

$$\langle v \rangle = \sqrt{\frac{8k_B T}{\pi m}}$$

3. Root-Mean-Square Speed (v_{rms})

The v_{rms} is perhaps the most physically significant speed because it relates directly to the **average kinetic energy** of the particles ($K = \frac{3}{2}k_BT$). It is defined as the square root of the average of the squared speeds:

$$v_{rms} = \sqrt{\int_0^\infty v^2 f(v)\, dv}$$

Using the integral form $\int_0^\infty x^4 e^{-ax^2} dx = \frac{3\sqrt{\pi}}{8a^{5/2}}$, the result simplifies beautifully to:

$$v_{rms} = \sqrt{\frac{3k_BT}{m}}$$

The Comparison: A Mathematical Hierarchy

Due to the "long tail" of high-energy particles, the distribution is skewed to the right. This creates a consistent mathematical hierarchy:

$$v_{mp} < \langle v \rangle < v_{rms}$$

$$\sqrt{2} < \sqrt{\frac{8}{\pi}} < \sqrt{3}$$

$$1.41 < 1.60 < 1.73$$

On the graph, v_{mp} is at the peak, $\langle v \rangle$ is slightly to the right, and v_{rms} is further still. This separation is the "signature" of statistical chaos—the fact that the most likely speed is not the one that carries the most energy.

9 Conclusion

This article just explores upon the dawn of statistical physics, which is an extravagant field of both mathematical and physical delight to the readers who seek statistical approval to the real-world phenomenons. There's much more to it than explored here. The classical statistical physics leads to the quantum statistical mechanics, which is even more amazing. For the readers, a few delightful and interesting problems have been included in this article too.

10 For the Curious

Statistical physics is best learned by getting your hands dirty with the combinatorics. Here are a few classical problems that demonstrate the "collision" of math and reality.

1. **The Drunkard's Walk (1D Random Walk):** A drunkard starts at $x = 0$ on a line. Each second, he takes a step of length L either to the left or right with equal probability ($p = q = 0.5$).
   - Show that his *average* position $\langle x \rangle$ after N steps is 0.
   - Show that his *root-mean-square* distance $\sqrt{\langle x^2 \rangle}$ is $L\sqrt{N}$.

- *Physics Link:* This explains why the smell of perfume spreads at a rate proportional to $\sqrt{t}$, not t.

2. **The Law of Sanity (Fluctuations):** If you have a room containing $N = 10^{24}$ molecules of air, calculate the probability that at any given moment, the number of molecules in the left half of the room differs from the number in the right half by more than 0.001%.

   - *Hint:* Use the Gaussian approximation of the binomial distribution. You will find the probability is so small it is effectively zero for the age of the universe.

3. **The Information of a Coin:** Show that the entropy of a single fair coin flip ($N = 1, \Omega = 2$) is exactly $k_B \ln 2$. If you have a gigabyte of "fair coins" (bits), what is the total physical entropy of that information?

4. **Gibbs Paradox (Mixing Entropy):** Suppose you have two identical chambers of the same gas at the same temperature. If you remove the barrier between them, the total multiplicity Ω technically increases. However, the entropy S does not change. Why?

   - *Research Topic:* Look up "Indistinguishability" and why we must divide the multiplicity by $N!$.